

Tracking Error as an Observation for Behavior Cloning on Low-Cost Arms

Luai Abuelsamen

University of California, Berkeley
luai_abuelsamen@berkeley.edu

Abstract: The difference between a commanded joint position and the position actually reached is available on any position-controlled robot that logs both, with no added sensor or fitted force model. We study this tracking error as an additional input to behavior cloning on a low-cost SO-101 arm. In simulation, placement success increases from 6.2% with position observations to 21.8% with tracking error across five training seeds, but position history reaches 20.3% across three seeds, leaving the specific benefit of tracking error unresolved. On hardware, policies trained on the same 50 demonstrations achieve 9/20 and 14/20 placements with tracking error against 5/20 and 1/20 for their position-only counterparts in two completed evaluations that begin after a scripted approach; a third evaluation was interrupted by hardware faults. The benefit varies substantially across training seeds and does not by itself show that the policy uses contact information. A study of 546 archived teleoperation episodes further shows that large tracking error at grasp onset does not predict grasp success.

Keywords: Behavior Cloning, Proprioception, Low-Cost Manipulation

1 Introduction

A position command describes an intended motion, and during manipulation the gap between that command and the position actually reached can persist for many steps—because a finger has met an object, because a joint is carrying a load, or simply because the actuator has not caught up. A policy given only the current joint position must infer all of this from its remaining observations, whereas supplying the previous command makes the discrepancy directly available.

For inexpensive robots this is attractive: it needs neither an added sensor nor a force-estimation model, and it can be reconstructed from existing demonstrations wherever the logged commands are the goals the servos actually received. Cheapness is not the question. Tracking error mixes contact with motion lag and actuator dynamics, and a history of positions may carry the same information. The question is whether exposing it helps an imitation policy, and what would justify attributing any gain to contact.

We run two experiments. A simulation grid sets explicit tracking error against action history, position history and a compensated residual under identical data and training budgets. A hardware study asks whether the simplest of these features helps ACT policies trained on 50 human demonstrations on one SO-101. Tracking error wins both completed hardware comparisons, by 4 and 13 placements out of 20, the second significantly. In simulation, position history matches it, so the results argue for keeping command history in policy interfaces without establishing that the policy reads contact.

2 Related work

Hwang et al. [1] estimate torque on RC servos by system identification; FACTR 2 [2] learns external-torque estimates from free-motion data and resamples training data by force. We use a simpler in-

put, the uncalibrated command–position difference, and make correspondingly weaker claims about force.

Wong et al. [3] show that leader–follower discrepancies in teleoperation can carry interaction cues: removing them from ACT’s action labels degrades performance in force-critical phases, while torque proxies restore contact-aware behavior. Our intervention instead adds the discrepancy to the policy observation, which makes this an observation-design study rather than a new sensorless-force principle. Adding the feature also adds access to the previous command. History that helps under partial observability can equally support a shortcut that copies the previous action instead of the action the task needs [4], so we compare several temporal observations rather than treating any gain over the base as evidence of force estimation.

3 Observation and experimental design

Let q_t be the measured joint position and g_{t-1} the preceding applied position goal, expressed in the same coordinate system. We define

$$\delta_t = g_{t-1} - q_t, \quad o_t^{\text{base}} = [I_t, \tilde{q}_t], \quad o_t^\delta = [I_t, \tilde{q}_t, \delta_t/c], \quad (1)$$

where I_t is the camera observation and $\tilde{q}_t$ uses training-set normalization. The scale $c = 8$ is fixed in each dataset’s joint-coordinate units and is not a conversion to torque. Under approximately static contact, tracking error may reflect load on a position servo, but during motion it also contains following error, so we use it as an observation without assigning every nonzero value to external contact.

Simulation. We train ACT policies [5] on 280 scripted pick-and-place demonstrations with randomized object and scene properties. All designs use the same data and training budget: 12,000 steps with 96-pixel front and gripper-camera images. Each trained policy is evaluated on 100 episodes with shared scene seeds across designs, with five training seeds for the original designs and three for the position-history control. The learning comparison uses placement success with the crush threshold disabled, so it tests grasping and transport rather than regulation within a safe force band.

Hardware. We collect 50 teleoperated demonstrations of placing an adapter inside a tape roll using one SO-101 and one overhead camera. Both policies use the same 18,549 training frames after removal of stationary prefixes, 96-pixel images, and 6,000 training steps. The proprioceptive input has six dimensions for the base and twelve for tracking error; all other architecture and optimization settings are shared. Three seeds are trained, one checkpoint per observation design and seed.

Evaluation begins after replaying 115 commands from a fixed demonstration, which places the arm near the grasp, after which each policy controls up to 400 steps. ACT predicts and executes 30-command chunks, so observations inform the next chunk rather than each command within it. A relative-target limiter is shared by both policies, and the tracking-error input uses the command returned *after* limiting. Trials are paired by training seed and physical reset, with base-first and tracking-error-first order alternating across pairs, and an operator records whether the adapter is placed inside the roll (Appendix A).

4 Results

In simulation, temporal information helps and its best representation is unresolved. Mean placement success is 6.2% for the position-only base, 13.8% with the previous action appended, and 21.8% with explicit tracking error. A compensated residual reaches 26.6%, while position history alone reaches 20.3% (Fig. 1). Tracking error exceeds the base by 15.6 points (Welch 95% CI 3.5 to 27.7) and position history by 1.5 points (CI −11.1 to 14.0); position history itself exceeds the base by 14.1 points (CI 4.5 to 23.8). Only the tracking-error and compensated-residual gains over the base clear the preregistered resolution rule of $15\sqrt{3/n}$ points for n seeds, and no contrast among the 28 design pairs survives a Holm adjustment (Appendix D). Because the residual subtracts a fitted

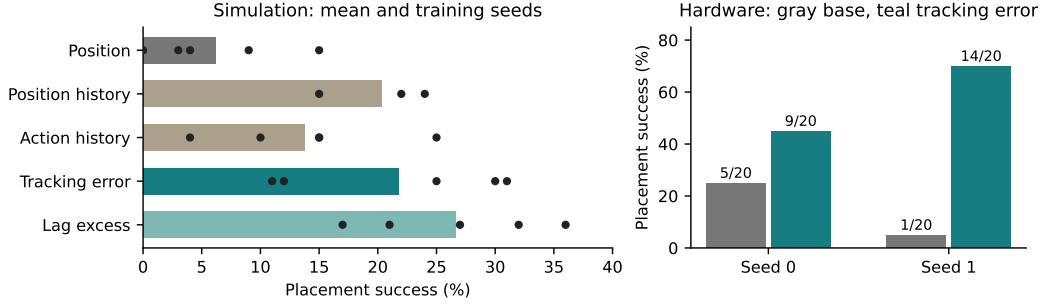


Figure 1: Placement success by observation design in simulation and in the two completed hardware evaluations. Left: bars are means; dots are individual training seeds (five per design, three for position history). Right: hardware success counts for the two completed evaluations. The full evaluation record, including an interrupted third comparison, appears in Appendix A.

Physical rollout: frames after policy handoff

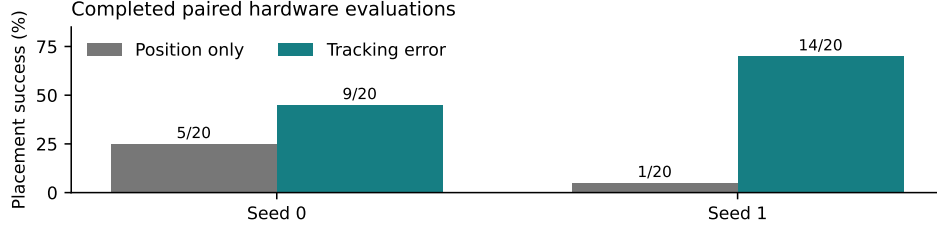


Figure 2: Top: three frames from a tracking-error policy (seed 1) after the scripted replay, from a showcase rollout that is not included in any count. Bottom: placement successes over 20 pairs for the two completed hardware evaluations.

free-motion term from tracking error and uses two previous commands, comparing it against one-command history changes both the information available and its representation. Position history performs comparably to tracking error without observing commands at all, so the grid supports supplying temporal information without resolving whether explicit tracking error is preferable to a suitable history. Predicting tracking error as an auxiliary training target gives 6.6%: this particular auxiliary objective does not recover the benefit of supplying it as input.

Hardware improvements depend on the trained policy. Table 1 reports each comparison separately: tracking error improves placement by 20 percentage points for seed 0 and 65 points for seed 1. Exact two-sided McNemar tests use only the discordant pairs. Seed 0 has six pairs favouring tracking error and two favouring the base ($p = 0.29$); seed 1 has fourteen and one ($p = 0.001$). The first comparison is inconclusive and the second is not. Both tests describe trial-to-trial variation for a fixed pair of checkpoints, not variation over training seeds. Pooling the completed seeds gives 20 discordant pairs favouring tracking error against 3 (stratified exact test $p = 0.0005$; conditional odds ratio 6.7, 95% CI 2.0 to 35), and the two discordance ratios are not detectably different (Fisher

Table 1: Hardware placements and paired outcomes for the two completed evaluations. Discordances list tracking-error-only / base-only successes. See Appendix A for the interrupted evaluation.

Seed	Evaluation	Base	Tracking error	Discordances	p
0	Complete	5/20	9/20	6 / 2	0.2891
1	Complete	1/20	14/20	14 / 1	0.0010

$p = 0.27$); these pooled figures still condition on two checkpoint pairs and are not an estimate over retraining. With 20 pairs the exact test has power 0.17 under seed 0’s observed discordance structure and 0.97 under seed 1’s, so the protocol could detect only large effects (Appendix A).

A third evaluation was interrupted by recurring hardware faults; its partial results and missing outcomes are reported in Appendix A.

A useful input is not necessarily an outcome label. We also test a preregistered prediction that larger tracking error at detected grasp onset accompanies successful grasps in archived teleoperation data. Sixteen datasets with 546 episodes pass the inclusion gates, of which only two meet the predicted positive effect-size threshold ($d > 0.5$); the pooled within-dataset effect is negative ($d = -0.39$), scored against a gripper-retention proxy, not manually verified placements. The result rejects this proposed labeling rule and illustrates why a discrepancy can indicate resistance without identifying a successful grasp (Appendix E).

5 Discussion and conclusion

Tracking error costs nothing to add: the command interface already carries both signals. In simulation and in two hardware comparisons it improved behavior cloning at a 50-demonstration budget. Whether the policy uses it as contact information is a separate question that this study leaves open. Position history is competitive in simulation, no history control has been run on hardware, and the hardware task never requires staying under a force threshold.

The two completed hardware seeds gave a 20-point and a 65-point improvement, and two seeds cannot bound that spread. The study uses one arm, one task and a fixed scripted approach, so it says nothing about full-task reliability, other objects or robots, or performance without the command limiter.

The next experiments are a position-history control on hardware, more training seeds per design, and a task with a measured force limit. Those three settle, respectively, whether tracking error adds information beyond history, how large the seed effect is, and whether a policy can use the signal to regulate contact.

References

- [1] Y. Hwang, Y. Minami, and M. Ishikawa. Virtual torque sensor for low-cost RC servo motors based on dynamic system identification utilizing parametric constraints. *Sensors*, 18(11):3856, 2018. URL <https://doi.org/10.3390/s18113856>.
- [2] S. Oh, J. J. Liu, T. Tao, P. Han, K. Shaw, S. Funabashi, R. Salakhutdinov, and D. Pathak. FACTR 2: Learning external force sensing for commodity robot arms improves policy learning. *arXiv preprint arXiv:2606.12406*, 2026. URL <https://arxiv.org/abs/2606.12406>.
- [3] K. H. Wong, L. Liu, and F. Dayoub. Beyond implicit force: Evaluating explicit force-torque proxies in action chunking with transformers. *arXiv preprint arXiv:2607.14578*, 2026. URL <https://arxiv.org/abs/2607.14578>.
- [4] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao. Fighting copycat agents in behavioral cloning from observation histories. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL <https://arxiv.org/abs/2010.14876>.
- [5] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Robotics: Science and Systems (RSS)*, 2023. URL <https://arxiv.org/abs/2304.13705>.

A Hardware protocol and complete evaluation record

Table 2: Full hardware evaluation record: success counts with Wilson 95% intervals in percentage points, discordant and concordant pair counts, exact McNemar p per seed, and the stratified exact test over the completed seeds. The interrupted seed has six recorded pairs, with 0/6 placements for each policy; its planned budget was 20 pairs. No confirmatory test is assigned to the incomplete comparison.

Seed	Evaluation	Base [95% CI]	Tracking error [95% CI]	TE / base only	both / neither	p
0	Complete	5/20 [+11, +47]	9/20 [+26, +66]	6 / 2	3 / 9	0.2891
1	Complete	1/20 [+1, +24]	14/20 [+48, +85]	14 / 1	0 / 5	0.0010
2	Interrupted	0/6 [+0, +39]	0/6 [+0, +39]	0 / 0	0 / 6	—
0+1	Stratified			20 / 3		0.0005

Table 3: Power of the exact two-sided McNemar test with 20 pairs at $\alpha = 0.05$, by the probability that a pair is discordant and the probability q that a discordant pair favours tracking error. Seed 0’s observed structure (8 discordant pairs, 6 favouring) gives 0.17; seed 1’s (15, 14) gives 0.97.

P(discordant)	$q = 0.7$	$q = 0.8$	$q = 0.9$	$q = 1.0$
0.2	0.02	0.05	0.10	0.20
0.4	0.11	0.26	0.53	0.87
0.6	0.19	0.46	0.80	1.00
0.8	0.29	0.64	0.93	1.00

Training. The recording contains 21,476 frames from 50 demonstrations at 30 Hz. Position and action statistics are computed on the recording; the training index then removes stationary prefixes, retaining 18,549 frames. Both policies use the same index and normalization, and recorded load channels are excluded from policy inputs. Gripper position is in the interface’s own units, not a calibrated force.

The ACT configuration uses a ResNet-18 image backbone, model width 256, four attention heads, feedforward width 1024, two encoder layers and one decoder layer. The action chunk and execution horizon are both 30 steps, with no temporal ensembling. Optimization uses AdamW at learning rate 10^{-4} , weight decay 10^{-4} , batch size 8 and gradient clipping at 10. All reported checkpoints are the final checkpoint at 6,000 steps. The observation projection necessarily changes width when six features are added; the remaining architecture and training settings are shared.

Execution and command semantics. Each rollout homes the arm to the mean demonstration start, replays the first 115 commands from demonstration 0, and hands control to the policy, so the reported task excludes autonomous approach from an arbitrary initial state. Each requested command passes through a relative-target limit of 8 in the interface’s joint units—a target-displacement bound rather than a force or velocity guarantee—and during both replay and policy execution the returned limited command supplies the next tracking-error calculation. The servo’s native goal quantization sits below this software interface. State and images are read at each control step, but ACT consults them only when its 30-command queue is empty.

An offline replay check reproduces the training and inference state vectors on all 21,476 recorded frames for the base, tracking error, action history, position history and compensated residual designs. In seed 0, the saved tracking-error observations after the first rollout step also reproduce from logged applied commands, which checks software consistency, not physical command-to-encoder latency. Collection-time command limiting has not been established from the available metadata, so archived action labels should be read as recorded goals, which need not equal the goals applied to a servo if collection used an additional limiter.

Pairing and inference. Hardware resets are manual, and within each seed consecutive trials form a pair with AB/BA order alternating between pairs. The operator knows the policy identity, so outcomes are not blinded. Seeds 1 and 2 were scheduled after observing seed 0, with their training

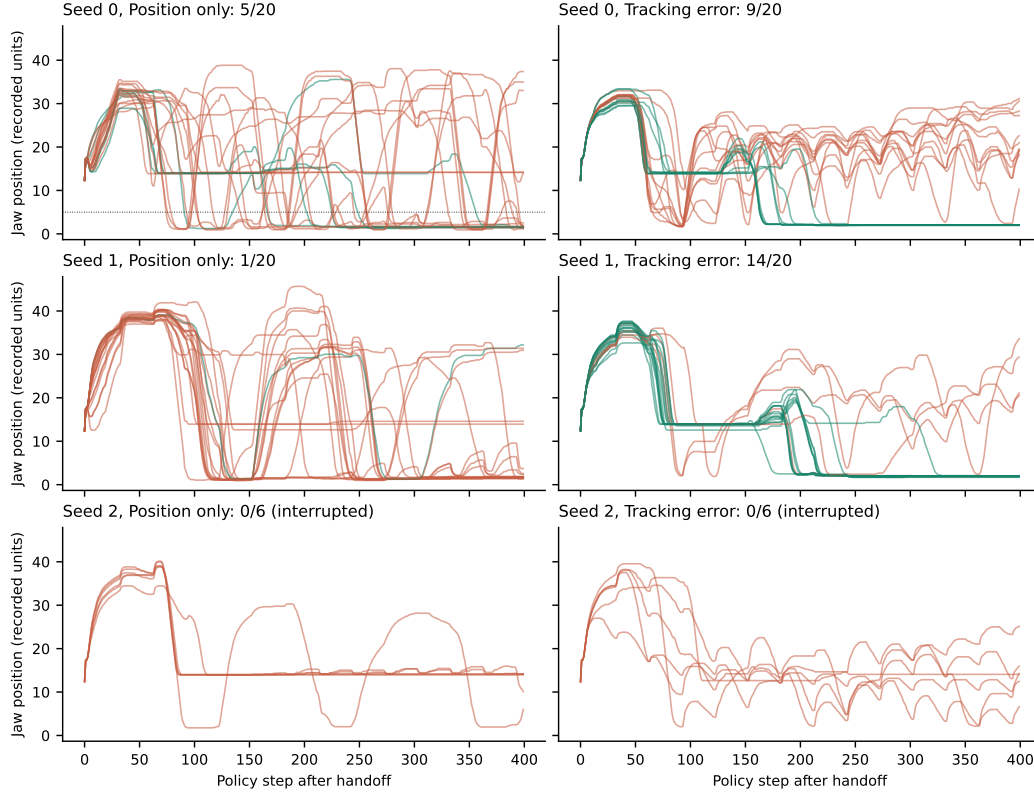


Figure 3: Recorded jaw position for every hardware rollout, by seed and policy; green traces placed the adapter, orange did not. Policy control begins at step 0 after the scripted replay. The dotted line marks the closure threshold used in Table 4.

recipe and evaluation budget fixed before their evaluations, and no policy was selected by hardware performance: every trained checkpoint was evaluated. The primary paired test is exact, two-sided McNemar with an improvement criterion of positive effect and $p < 0.05$; for the completed seeds, paired percentile bootstrap intervals (20,000 pair resamples, random seed 0) are $[-5, +45]$ and $[+40, +90]$ points, describing trial uncertainty for fixed checkpoints, not confidence intervals over retraining.

Interruptions and missing data. Seed 0 has 40 outcomes and 40 trajectories. Seed 1 has 40 outcomes and 39 trajectories: two rollouts, one per policy, completed their control horizon but reported a gripper overload while torque was being disabled; both are operator-labeled failures, and one trajectory file was lost to a logging bug that was fixed before evaluation resumed. Seed 2 was stopped after twelve labeled rollouts, all failures, forming six pairs; three reported the same overload at shutdown, and a thirteenth rollout completed its horizon but its outcome was not entered. We leave that outcome missing rather than infer it from joint motion, and we have no diagnosis of the recurring overload. In the seed-2 rollouts the jaw never closed in five of six position-only and three of six tracking-error trials, against three of 20 and one of 20 in seed 0 and two of 20 and none of 19 in seed 1 (Table 4), which is consistent with a gripper fault rather than a policy difference, although the traces cannot prove it. Trial-level detail is in the repository’s evaluation records.

B Exploratory trajectory analysis

Every recorded rollout was classified from its jaw trace as never closed, closed then reopened, or closed and held (Table 4, Fig. 3). All 23 tracking-error successes in seeds 0 and 1 closed once and held to the end of the horizon, and 15 of their 16 failures either never closed or reopened. Position-

Table 4: Jaw-trace classes for all recorded rollouts. Never closed: jaw never below 5 recorded units. Reopened: an excursion above 8 units after closing. Closure step is the median first step below the threshold. One seed-1 tracking-error trajectory is missing.

Seed	Policy	Outcome	n	never closed	closed, reopened	closed, held	median step of closure
0	Position only	success	5	0	1	4	203
0	Position only	failure	15	3	8	4	110
0	Tracking error	success	9	0	0	9	165
0	Tracking error	failure	11	1	10	0	83
1	Position only	success	1	0	1	0	127
1	Position only	failure	19	2	13	4	116
1	Tracking error	success	14	0	0	14	206
1	Tracking error	failure	5	0	4	1	115
2	Position only	failure	6	5	1	0	88
2	Tracking error	failure	6	3	3	0	241

only successes are too few to characterize. Successful tracking-error rollouts closed later, at median step 165 and 206 against 83 and 115 for failures, and in seed 0 they requested smaller moves: the median fraction of steps altered by the command limiter is 0.09 for tracking error and 0.32 for the base, while in seed 1 the fractions are 0.32 and 0.35. Over the last 100 steps of the seed-0 tracking-error rollouts, median summed joint-coordinate travel, a descriptive sum over mixed coordinates, is 14.3 for successes and 93.6 for failures. Later closure and less motion in successful trials could simply follow from progressing further through the task. Traces were grouped by outcome after the fact and contain no video, object pose or force, so they cannot say whether a failure was a missed grasp, a slip or a misplacement, and none of these regularities is a validated success detector.

C Measurement context

A separate static experiment on one shoulder-lift STS3215 uses twelve hanging masses at five poses (60 measurements). Across those measurements, the fitted relationship between the load register and tracking error is $\text{Present_Load} = -4.70\delta + 0.1$, with $R^2 = 0.976$. Its relation to applied mass is far less stable, with per-pose R^2 for tracking error ranging from 0.085 to 0.910, so the tight load-register association does not establish an equally accurate physical-load measurement.

In the 50 teleoperation demonstrations, the gripper load register reaches 500 in every episode and remains there in 16.5% of frames, while tracking error continues to vary across those same frames (Fig. 4); over all frames the two channels correlate at $r = -0.92$. The inspected follower configuration sets a reduced gripper torque limit, consistent with that observed plateau. This motivates inspecting the command-position discrepancy when telemetry saturates, but it does not calibrate the remaining discrepancy in newtons: no jaw load-cell calibration is available, and simulator force conversions such as the fitted 2.00 N/count slope and 4.3 N held-out RMSE over 4–78 N describe the simulated plant only.

D Additional simulation and hardware comparisons

Simulation recordings use integer encoder counts at 15 Hz, obtained by subsampling the 30 Hz controller. The scripted teacher regulates the jaw using tracking error, and the successful demonstration set includes randomized grip depth and recovery perturbations. These collection choices are shared across observation designs, of which the position-history control observes $[q_t, q_{t-8}]$ and one-command history observes $[q_t, g_{t-1}]$. Neither comparison makes the simulation and hardware tasks identical: camera views, demonstrations and recording cadence all differ between the two settings.

The original simulation study includes an auxiliary tracking-error target and a latched seat token in addition to the designs in Fig. 1, with mean placement rates of 6.6% and 21.8% across five seeds; the seat token matches tracking error exactly. The compensated residual is $e_t = \delta_t - \hat{K}(g_{t-1} - g_{t-2})$, where each joint’s coefficient is fitted once on presumed free-motion demonstration frames.

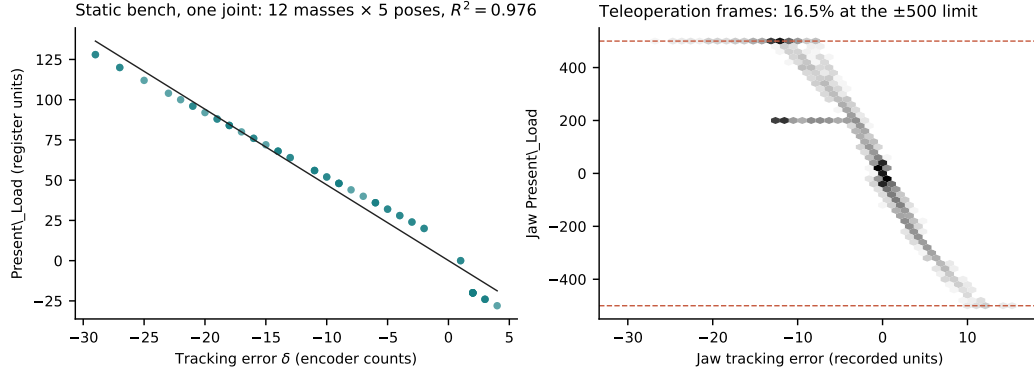


Figure 4: Left: static bench, one shoulder-lift servo, load register against tracking error over twelve hanging masses at five poses, with the fitted line. Right: jaw load register against jaw tracking error on the teleoperation frames (log-count shading); the dashed lines are the configured ± 500 torque limit.

A two-command history control reaches 14.7% over three seeds; comparing it with the residual holds available raw information fixed but does not match input width. Under the original study’s preregistered practical resolution rule, $15\sqrt{3/n}$ percentage points for n training seeds, neither this comparison nor the residual–position–history comparison resolves at three seeds, and the stronger feature-specific interpretation remains open. Table 5 lists the main design contrasts with Welch intervals; the Holm column adjusts across all 28 pairwise comparisons.

Table 5: Simulation design contrasts in placement-success points, with Welch 95% intervals, unadjusted and Holm-adjusted p values across all 28 pairs, the preregistered resolution rule $15\sqrt{3/n}$ for the smaller seed count, and whether the difference clears it.

Contrast	Points	Welch 95% CI	p	Holm p	Rule	Clears
Position history – Position	+14.1	[+4.5, +23.8]	0.013	0.33	15.0	no
Action history – Position	+7.6	[-2.5, +17.7]	0.121	1.00	11.6	no
Tracking error – Position	+15.6	[+3.5, +27.7]	0.019	0.44	11.6	yes
Compensated residual – Position	+20.4	[+10.2, +30.6]	0.002	0.05	11.6	yes
Auxiliary target – Position	+0.4	[-7.1, +7.9]	0.903	1.00	11.6	no
Tracking error – Position history	+1.5	[-11.1, +14.0]	0.784	1.00	15.0	no
Compensated residual – Position history	+6.3	[-4.6, +17.1]	0.206	1.00	15.0	no
Tracking error – Action history	+8.0	[-4.9, +20.9]	0.188	1.00	11.6	no
Compensated residual – Two-command history	+11.9	[+2.2, +21.7]	0.025	0.49	15.0	no
Compensated residual – Tracking error	+4.8	[-8.1, +17.7]	0.413	1.00	11.6	no

An earlier hardware comparison, run before the tracking-error study, evaluated the compensated residual: the base placed 7/10 objects and the residual 0/10. The coefficients fitted from these real demonstrations differ markedly from the simulation fits, and a free-motion labeling heuristic includes contact frames. Both are plausible sources of mismatch, not a diagnosis. Coefficients above one do not by themselves invalidate the regression, and that runner also fed back requested rather than post-limit commands, so preprocessing and execution both differ from the tracking-error study.

The simulation task admits a crush threshold, but the learning grid does not activate it. A post-hoc screen of existing checkpoints at 120 and 60 N produces heavy crushing and near-floor success, failing the predicted advantage of the compensated residual. A separate runtime guard reduces crush events from 85 to 3 across 180 crush-tier episodes, while eliminating all 12 unguarded placements across its broader 270-episode comparison. Neither result is learned force regulation, and the guard was not used on hardware.

Larger-budget experiments do not establish a scaling advantage. An earlier 600-demonstration comparison of tracking error and one-command history reaches 57% and 60%, respectively, over two seeds at 224 pixels and 50,000 steps. A later 600-demonstration screen saved at 26,000 steps contains three base and three residual evaluations, each with 0/100 placements. These floor results are reported separately from the 280-demonstration grid; they support neither a positive scaling claim nor equivalence between observations.

E Archived teleoperation study

The preregistered corpus prediction was that tracking error at a detected seat event would be larger in proxy-successful episodes, with Cohen’s $d > 0.5$ in at least ten of twenty targeted datasets. Sixteen datasets and 546 episodes pass the frozen gates, including 283 proxy successes under a success proxy of continued gripper closure through transport. Only two datasets exceed the positive threshold, and the pooled within-dataset effect is $d = -0.39$ —a failure of the proposed direction and labeling rule, though not proof that no function of command and position history can predict outcome. Eleven of the sixteen point estimates are negative and seven of those intervals exclude zero; two are positive with intervals excluding zero (Fig. 5).

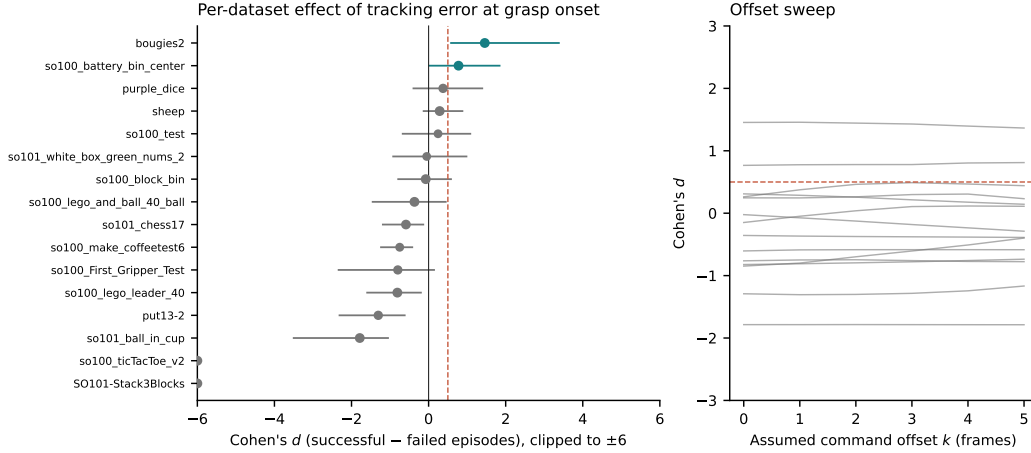


Figure 5: Left: per-dataset Cohen’s d of tracking error at grasp onset between proxy-successful and failed episodes, with 95% intervals, clipped to ± 6 ; marker area scales with episode count and the dashed line is the preregistered $d = 0.5$ threshold. Right: the same effect sizes under assumed command offsets of zero to five frames.

Empty closure against a mechanical stop and closure on an object can both produce a seat event, and a large command discrepancy can equally reflect an operator continuing to close against resistance. These ambiguities explain why a contact-related feature need not be a success label, although the corpus does not isolate their causal contributions. Recorded leader commands may differ from applied follower goals and the timing is only approximately aligned, but an offset sweep from zero to five recorded frames leaves the qualitative conclusion unchanged, with median per-dataset effect-size shift 0.12.